LAB WORKSHEET

Physical Topology Lab

Design a physical network: rooms, cabinets, structured cabling, labels and IP addressing.

Structured cabling

Device → patch lead → wall outlet → horizontal cable → patch panel → patch cord → switch. Cabinets are measured in rack units (1U = 44.45 mm); protect active kit with a UPS.

Key rules

- Copper: 90 m permanent link, 100 m channel. Fibre: OM3 ≈ 300 m, OM4 ≈ 400 m (10 Gb/s), OS2 10 km+.
- Access points, IP phones and CCTV cameras need a **PoE** switch port. A console (rollover) cable is for configuration only.
- Hosts never use the network or broadcast address; no two devices share an IP.
- A host's default gateway is the router interface in its own subnet (usually .1). Router-to-router links share one subnet (often /30).
- Use private ranges inside the LAN; only the ISP-facing interface is public (203.0.113.0/24 in this lab).

Controls

Drag from the Library · drop rack kit onto a cabinet · double-click a cabinet for its rack elevation · L = Link tool · double-click a label to edit · F7 Check design · F8 Cable schedule · F9 Addressing table · Subnet helper + Auto-address.

Tasks

Guided: 15 objectives from the comms room to a fibre riser and full addressing. Independent: Calder Valley Dental Practice (two floors, annex, staff LAN 10.20.10.0/24, CCTV VLAN 30). Free practice: small office, two-site network and school IT suite design tasks.

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